

Test Design

Terrain-based scoring, route deviation and hill-route regression tests

Project	Course	Case Study	Version
HaulTrack - 1,800-truck fleet telematics	B.Tech CSE - Software Engineering & Project Management	No. 19	1.0 (baseline)

1. Test strategy

The suite is built so it **fails the previous vendor's algorithm**. Expected values below are computed from the scoring model in the Design Pack (Score = 100 - (B + S + I); B = min(50, 5h)); the "Legacy" column re-scores the same inputs with a single flat 3.0 m/s² threshold and no gradient input, which reproduces the reported behaviour of the abandoned vendor (penalising unavoidable hill braking). Where legacy and new scores differ by more than 2 points, the legacy model fails that test.

Level	What is tested	Tool / approach	Exit criterion
Unit	Threshold lookup, band boundaries, penalty caps, exposure rules	Automated, table-driven	100% of listed cases pass; branch coverage >= 90% on scoring and deviation code
Integration	Enrichment -> event detector -> scoring	Replay of synthetic trip files	Expected scores match to 0.1 point
Fairness / statistical	TN-3, TN-4, TN-6 on a synthetic and (when available) pilot cohort	Cohort simulation, paired comparison	Spearman rho <= 0.10; paired gap <= 2.0 points
Load	1,800 trucks, 60 records/s average, 180/s peak	Synthetic generator	No loss; end-to-end latency within target
Policy enforcement	Access and export controls from Deliverable 06	Role-based negative tests	All prohibited actions blocked and logged
UAT / shadow pilot	Real routes at two depots, no consequences for drivers	Pilot reviewed by Governance Board	TN-7 release gate met

2. Terrain-based scoring tests

Thresholds (m/s² net deceleration, downhill bands): under 2% = 3.0; 2 to under 5% = 3.3; 5 to under 8% = 3.6; 8% and above = 4.0; flat and uphill = 3.0. Speeding and idling penalties are zero in these cases to isolate braking.

ID	Case	Distance	h per 100 km (new)	New score	Legacy score	Legacy vs new
ST-01	Flat road, no harsh events	100 km	0.0	100.0	100.0	same
ST-02	Flat, 4 events at 3.4 m/s ²	100 km	4.0	80.0	80.0	same
ST-03a	Boundary: flat, 1 event exactly 3.00 (counts, >=)	100 km	1.0	95.0	95.0	same
ST-03b	Boundary: flat, 1 event 2.99 (does not count)	100 km	0.0	100.0	100.0	same
ST-04	Rolling 3% descent, 5 events at 3.2 (below 3.3)	100 km	0.0	100.0	75.0	FAIL
ST-05	Hilly 6% descent, 8 events at 3.5 (below 3.6)	100 km	0.0	100.0	60.0	FAIL
ST-06	Mountain 9% descent, 10 events at 3.9 (below 4.0)	50 km	0.0	100.0	50	FAIL
ST-07	Mountain 9%, 2 genuinely harsh events at 4.6	50 km	4.0	80.0	80.0	same
ST-08	Grade boundary: exactly 5.0% uses the 5-8% band (3.6); 1 event 3.5	100 km	0.0	100.0	95.0	FAIL
ST-09	Uphill +6%: no relief; 1 event at 3.2 counts	100 km	1.0	95.0	95.0	same
ST-10	Cap: flat, 15 events at 3.5 (B capped at 50)	100 km	15.0	50	50	same
ST-11	Mixed 100 km: 60 km flat 1 event 3.4; 40 km mountain 5 events 3.8	100 km	1.0	95.0	70.0	FAIL

Reading the table. ST-04 to ST-06 and ST-11 are the terrain cases: the same driver behaviour that earns a good score under the new rules is penalised heavily by the legacy rule. ST-07 and ST-09 prove the fix is not blanket

leniency: genuinely harsh braking, and braking uphill, is still penalised identically by both. ST-03/ST-08 test boundaries so a band or threshold cannot drift by one step.

ID	Case	Expected result
ST-12	300 km in the 28-day window	Status NOT_RATED; no score shown or exported
ST-13	Unknown gradient on 8% of the distance	Score withheld (limit 5%); data gap flagged, driver not penalised
ST-14	Cohort: 200 synthetic drivers with identical behaviour distribution, half on flat routes, half on mountain routes	Spearman $ \rho $ between hilliness and score ≤ 0.10 (TN-3); mean gap ≤ 2.0 points
ST-15	Paired: 30+ drivers who drive both flat and hill routes with unchanged behaviour	Per-driver score difference ≤ 2.0 points (TN-4)
ST-16	Driver is moved from flat to a hill route	Score change ≤ 2.0 points; no action permitted for 28 days (TN-6)
ST-17	Elevation data accuracy check on surveyed segments	RMSE ≤ 1.0 percentage point (TN-1)

3. Hill-route regression cases (release blockers)

These reproduce the abandoned vendor's failure. They run on every build; a failure blocks release (TN-7).

ID	Scenario	Inputs	Assertion	New	Legacy	Legacy outcome
HR-01	Ghat descent: careful driver	60 km, 9% descent, 14 service-brake applications; 11 reach 3.0-3.9 m/s ² , none above 3.9	Score ≥ 95	100.0	50	FAIL
HR-02	Same descent, one real harsh event	As HR-01 plus 1 event at 4.5 m/s ²	Score < 100 and ≥ 85 (penalised, not wiped out)	91.7	50	FAIL
HR-03	Hilly 6% route, steady driver	100 km, 6% descents, 8 events at 3.5	Score ≥ 95	100.0	60.0	FAIL

ID	Scenario	Assertion
HR-04	Paired: same driver, same careful behaviour, flat 60 km route vs HR-01 route	Scores differ by ≤ 2.0 points (legacy gap is about 50 points)
HR-05	Replay of the previous vendor's own dataset, if obtainable (otherwise the synthetic suite stands in)	Median score of hill-route drivers within 3 points of flat-route drivers

Result summary. Under the legacy algorithm HR-01, HR-02 and HR-03 all fail (scores of 50-60 for careful hill driving). The new algorithm passes all three and still penalises the genuinely harsh event in HR-02.

4. Route deviation tests

Defaults: corridor 500 m, 3 consecutive off-corridor reports (90 s) to raise a NOTICE, escalate to ALERT at 10 min or 2 km, GPS must meet the HDOP limit.

ID	Scenario	Expected result
RD-01	Truck follows planned route	No alert
RD-02	GPS jitter of 60 m, still inside corridor	No alert
RD-03	One outlier fix 2 km away, then back on route	No alert (counter resets)
RD-04	Two consecutive off-corridor reports (60 s)	No alert yet
RD-05	Three consecutive off-corridor reports (90 s)	NOTICE raised at the third report exactly
RD-06	Distance exactly 500 m vs 501 m	500 m inside (no count); 501 m outside (counts)
RD-07	Off route and persisting 10 min	Escalated to ALERT
RD-08	Fast departure 2.1 km off route in 4 min	Escalated to ALERT (distance rule)
RD-09	Road-closure feed active on the corridor	Suppressed; reason code logged
RD-10	Inside depot geofence or planned stop	Suppressed; no notice
RD-11	Five off-route reports with poor GPS quality	No alert; state held; low-quality fixes logged

ID	Scenario	Expected result
RD-12	Tunnel: no reports for 2 min, then back on route	No alert (missing data is not deviation)
RD-13	Dispatcher approves a reroute	New corridor applied; no further alert on it
RD-14	Truck returns to corridor	Open deviation auto-clears
RD-15	Duplicate and out-of-order records	Ignored; no double counting
RD-16	Deviation history vs safety score inputs	Deviation data absent from every score calculation (D6)

5. Policy enforcement tests

ID	Scenario	Expected result
PE-01	Manager attempts to export scores as a disciplinary list	Blocked; attempt logged and reported to the Governance Board
PE-02	Insurer API called with a driver identifier	Rejected; only aggregates and pseudonymised events served
PE-03	Ignition off or driver logged off	No location record stored
PE-04	Raw record older than 90 days	Purged by the retention job; purge logged
PE-05	Driver opens own score	Full breakdown with events, gradient band and threshold shown
PE-06	Score under appeal	Marked "under review"; not usable for any decision until resolved

6. Traceability

Requirement	Covered by
TN-1	ST-17
TN-2	ST-03 to ST-11, HR-01 to HR-03
TN-3	ST-14
TN-4	ST-15, HR-04
TN-5	ST-07, HR-02
TN-6	ST-16
TN-7	HR-01 to HR-05 as release gate
D6 (deviation separate)	RD-16
D3 / D5 policy decisions	PE-01 to PE-06